

Title

A Prospective Evaluation of Integrated Online Care Coordination for Managing Pruritis and Sleep Disruption in Atopic Dermatitis: Study Design and Methods

Abstract

Atopic dermatitis remains a chronic inflammatory condition with significant morbidity, particularly regarding itch severity and secondary sleep disturbance. Coordinated multidisciplinary care delivery via digital platforms may offer improved accessibility and outcomes. This randomized trial will enroll approximately 100 adult participants with clinically confirmed atopic dermatitis from dermatology clinics. Participants will be randomized 1:1 to receive either integrated online care coordination involving clinical assessment, treatment optimization, and supportive messaging, or continued standard routine care. The online platform will include specialist nurse support, treatment adherence monitoring, and patient education modules. Primary outcome is objective and subjective measures of pruritis intensity at 12 weeks. Secondary outcomes include standardized sleep quality assessments, disease severity scoring, and dermatitis-specific quality-of-life instruments. Baseline disease severity, comorbidities, and psychosocial factors will be documented. Data collection will occur at baseline and 12 weeks post-randomization. Analysis will employ mixed-model intent-to-treat methodology.